



AIRLINE LOYALTY CUSTOMER CHURN PREDICTION & CUSTOMER ANALYTICS DASHBOARD

*End-to-End Machine Learning & Business Intelligence Project
Project Report & Technical Documentation*

16,737

Unique Customers

389,065

Records Analysed

12.4%

Churn Rate

0.981

Model ROC-AUC

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Technologies: Python · Pandas · NumPy · Matplotlib · Seaborn · Plotly · Scikit-Learn ·
XGBoost · LightGBM · Streamlit · Git & GitHub

Dataset Period: January 2017 — December 2018

Executive Summary

The airline industry depends heavily on loyal customers who generate recurring revenue through repeat bookings, points redemption, and premium service upgrades. When these customers leave a loyalty programme, airlines not only lose immediate revenue but also incur the higher cost of acquiring new customers to replace them. This is the business problem at the heart of this project.

This project delivers a complete, end-to-end data science solution — from raw data to a live web-based dashboard — that helps airline business teams identify which customers are likely to churn, understand why they are churning, and take proactive steps to retain them.

The project was completed in the following stages:

- Data Collection and Integration — merging customer demographics, flight activity, and loyalty data
- Data Cleaning and Preprocessing — handling missing values, duplicates, and type corrections
- Exploratory Data Analysis — discovering churn patterns, customer behaviour, and key business drivers
- Feature Engineering — creating meaningful variables that capture customer engagement and loyalty
- Machine Learning Model Development — training and comparing seven classification algorithms
- Model Evaluation and Selection — choosing the best model based on business-relevant metrics
- Dashboard Development — building an interactive Streamlit web application for non-technical users

The final solution includes three core deliverables:

1. A Customer Churn Prediction Model with 97.1% accuracy and 0.981 ROC-AUC, capable of scoring all 16,737 customers with a personalised churn probability
2. A Web-Based Interactive Dashboard with five pages covering executive KPIs, customer demographics, loyalty programme health, revenue insights, and live churn prediction
3. Business Recommendations for Customer Retention — targeting the 2,419 high-risk customers identified by the model



Bottom Line

This project transforms raw airline loyalty data into actionable business intelligence. The churn model correctly identifies over 80% of customers who are actually at risk of leaving — giving the airline's retention team a reliable, prioritised list to act on.

1. Problem Statement

Airlines invest millions of dollars in customer acquisition — advertising campaigns, sign-up bonuses, onboarding offers — yet many of these customers disengage or cancel their loyalty membership within the first few years. Each lost loyalty member represents not just one cancelled flight, but the loss of a

long-term revenue relationship that could have spanned years of bookings, upgrades, and points spending.

The business problem addressed by this project is straightforward:

? Core Problem Statement

"Predict which customers are most likely to churn from the airline loyalty programme and identify the key factors driving their decision to leave — so that the business can intervene early with personalised retention strategies."

Business Objectives

- Identify customers with high churn risk before they actually leave
- Understand the behavioural and demographic factors that predict churn
- Enable the marketing team to design targeted, personalised retention campaigns
- Increase customer lifetime value by reducing avoidable attrition
- Provide a self-service dashboard so business teams can monitor and act without needing data science support

The solution needed to be practical and accessible. Rather than producing a model that only data scientists could use, the goal was to package the predictions into a user-friendly web application that a marketing manager or customer success team could open on their browser and immediately act on.

2. Dataset Description

The project uses three datasets from a Canadian airline loyalty programme, covering the period January 2017 through December 2018. These were integrated into a single unified analytical dataset for modelling and dashboard development.

2.1 Customer Loyalty History

This dataset contains the core demographic and programme information for every loyalty member. Each row represents one unique customer.

Field	Description
Customer ID / Loyalty Number	Unique identifier for each loyalty member
Country / Province / City	Customer geographic location
Gender	Customer gender
Education	Highest level of education completed
Marital Status	Single, Married, or Divorced
Salary	Annual salary bracket
Loyalty Card	Tier of loyalty membership — Aurora, Nova, or Star

CLV	Customer Lifetime Value — estimated total revenue from the customer
Enrollment Year / Month	When the customer joined the loyalty programme
Cancelled	Target variable — 1 if the customer churned, 0 if retained

2.2 Customer Flight Activity

This dataset contains monthly transaction-level records. Each row represents one customer in one calendar month, capturing their flight and points activity for that period.

Field	Description
Total Flights	Number of flights taken in that month
Distance	Total kilometres flown
Points Accumulated	Loyalty points earned through flights and partner activity
Points Redeemed	Points spent on rewards, upgrades, or redemptions
Dollar Cost Points Redeemed	Monetary value of points redeemed (in CAD)

2.3 Dataset Integration

The three datasets were merged on Customer ID and calendar keys (Year, Month) to produce a single analytical dataset with 389,065 rows and 46 engineered features. This integrated view allows the model to consider both who the customer is (demographics, tier, CLV) and what they do (flight activity, points behaviour) simultaneously.

389,065 Total Records	16,737 Unique Customers	2 Years Data Coverage	46 Features (Final)
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3. Data Preprocessing & Cleaning

Before any analysis or modelling can begin, the raw data must be cleaned and prepared. Poor data quality is one of the most common reasons machine learning models fail in production. This section describes every step taken to ensure the data was accurate, consistent, and ready for analysis.

3.1 Data Inspection

The first step was to understand the shape and structure of each dataset. This involved:

- Checking the dimensions (rows and columns) of each file
- Reviewing column names and data types to catch misclassifications
- Running statistical summaries (mean, median, min, max, standard deviation) to detect anomalies

- Identifying columns with missing or null values and quantifying the extent of missingness
- Checking for duplicate records that could bias model training

3.2 Missing Value Treatment

Missing data is common in real-world loyalty datasets — customers may not disclose salary, or flight activity may be zero in months they did not fly. The approach taken was:

- Numeric columns (salary, CLV, flight metrics) — imputed using the median value during model preprocessing to avoid the influence of outliers on the imputed value
- Categorical columns (province, education, marital status) — imputed using the most frequent value (mode)
- Records with missing target variable values were removed, as they cannot contribute to supervised learning

Why Median Imputation?

The median is more robust to outliers than the mean. In a salary distribution where some customers earn significantly more than others, using the median as the fill value produces a more representative result than the mean, which would be skewed by extreme values.

3.3 Duplicate Handling

Duplicate records were identified at the customer-month level. Removing duplicates is critical because they artificially inflate sample sizes and can cause a model to overfit — memorising repeated records rather than learning genuine patterns.

3.4 Data Type Corrections

Several columns required type conversion to ensure they were processed correctly by downstream code:

- Enrollment Year and Month were treated as numeric features, not datetime objects, to capture the linear relationship between tenure and churn
- Loyalty Card tier (Aurora, Nova, Star) was encoded as an ordinal categorical variable
- Boolean flags created during feature engineering were converted from True/False to 1/0 integers

3.5 Dataset Merging

The three source datasets were merged using pandas merge operations on Customer ID and calendar keys. The resulting unified dataset was validated to confirm row counts matched expectations and no customers were lost during the join. This single analytical dataset became the foundation for all subsequent analysis and modelling.

4. Exploratory Data Analysis (EDA)

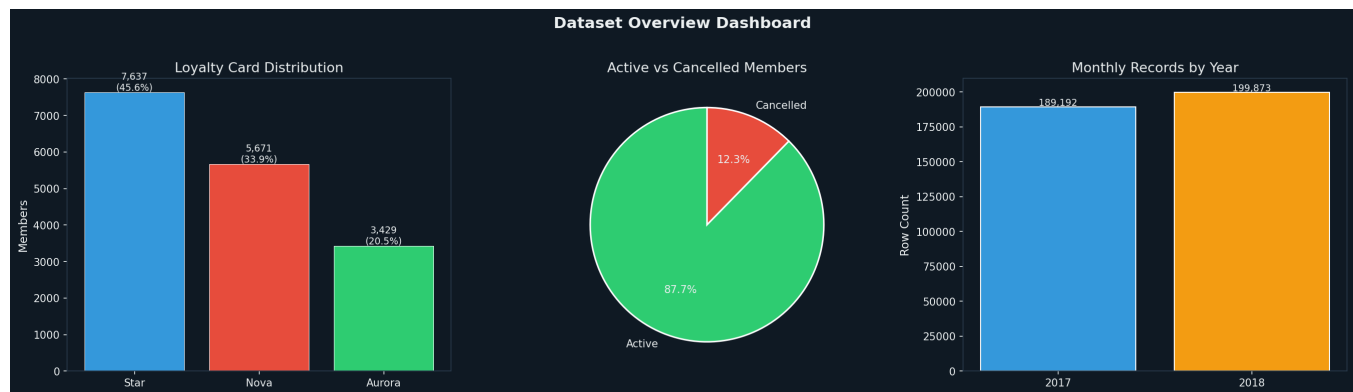
Exploratory Data Analysis is the investigative phase of the project — the stage where we ask questions of the data and let it reveal its patterns before building any model. EDA prevents the common mistake of jumping straight into modelling without understanding what the data is actually telling us.

4.1 Univariate Analysis — Understanding Individual Variables

Univariate analysis examines one variable at a time to understand its distribution and composition.

Loyalty Card Distribution

The customer base is divided across three tiers — Star (base), Nova (mid-tier), and Aurora (premium). Star members make up the largest group at 45.6% of customers, while Aurora members are the smallest and most valuable segment at 20.5%.

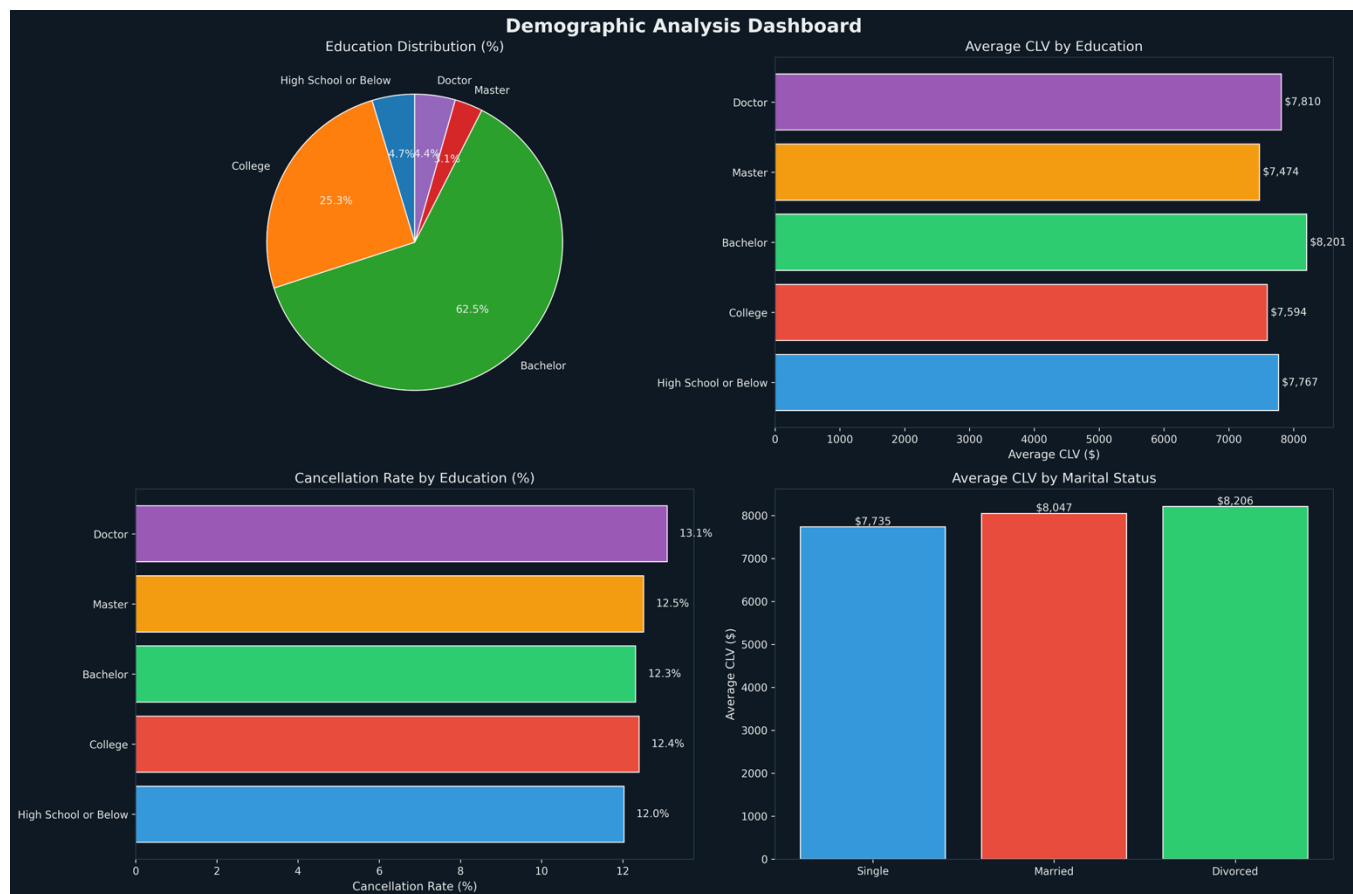


Salary Distribution

Customer salaries range from below \$40,000 to above \$150,000 annually. The distribution shows a moderate right skew — most customers fall in the \$40,000–\$100,000 range, with a smaller proportion of high earners in the Aurora tier.

Education and Marital Status

The majority of loyalty members hold a Bachelor's degree or higher. Married customers represent the largest demographic group, followed by single customers. These patterns matter because education and marital status correlate with flight frequency and points engagement.



4.2 Bivariate Analysis — Churn Drivers

Bivariate analysis compares churn status (churned vs retained) against individual features to identify which factors separate loyal customers from those who leave.

Churn vs Loyalty Card Type

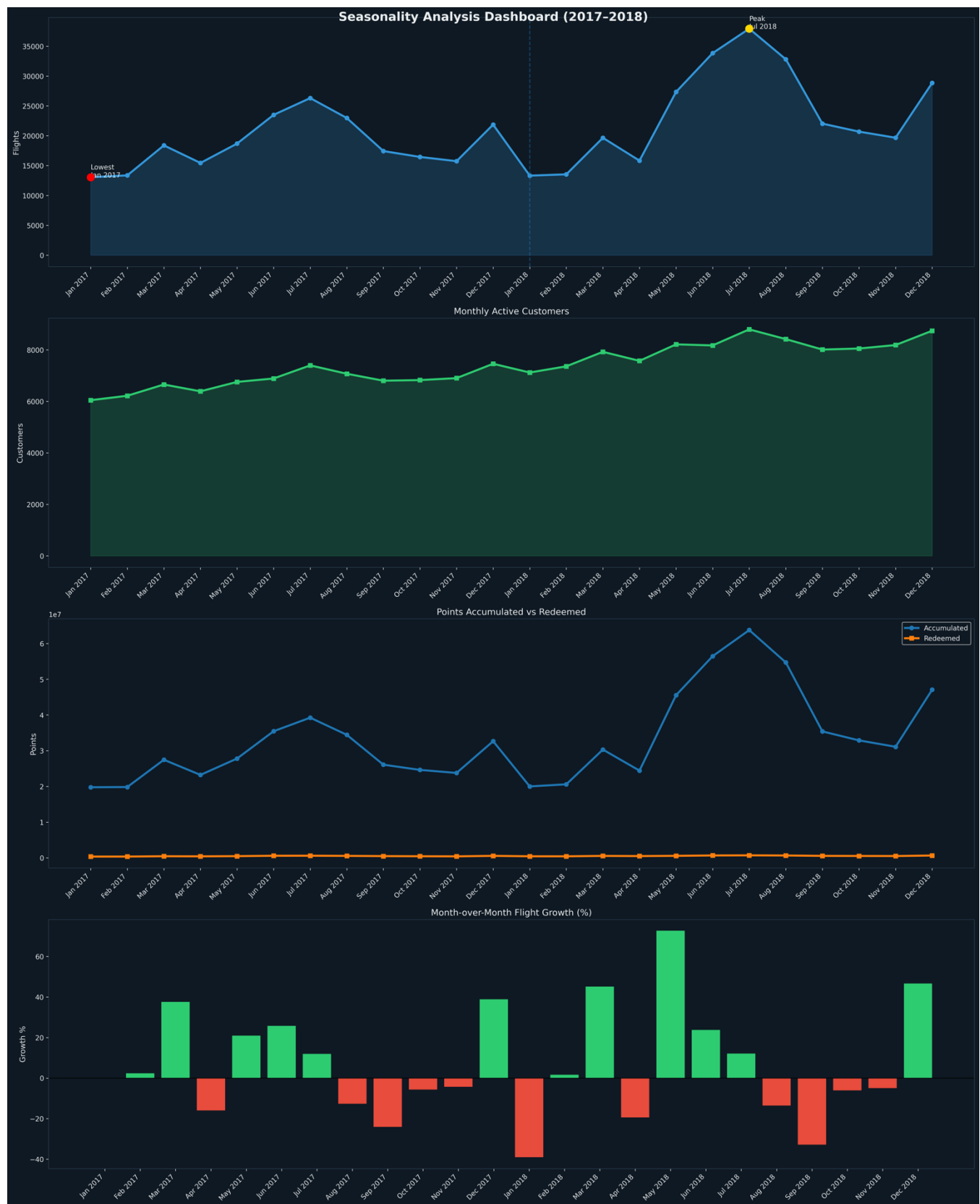
Aurora members, despite being the most valuable customers, have a slightly higher churn rate (13.1%) than Star members (11.8%). This counterintuitive finding suggests that premium customers may have higher expectations and are more likely to leave if those expectations are not met — making Aurora the highest-priority segment for personalised retention efforts.

Churn vs Flight Activity

Customers with fewer total flights show significantly higher churn rates. This is intuitive — customers who fly frequently are more engaged with the programme and have more invested in their points balance. Low flight activity is one of the clearest early warning signs of an at-risk customer.

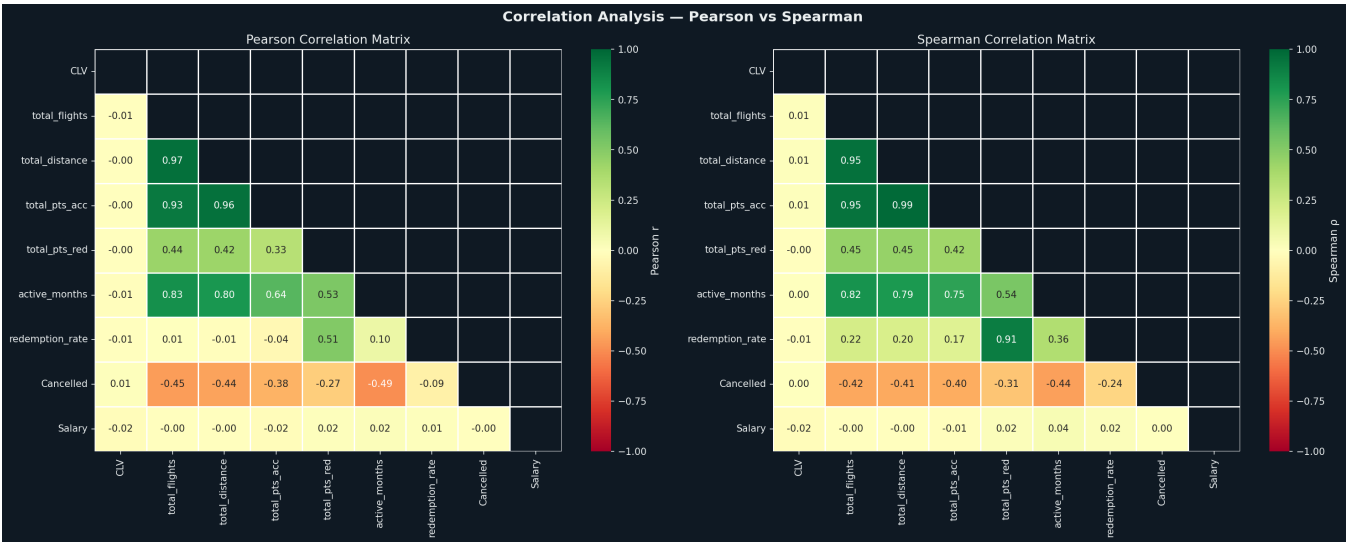
Churn vs Points Redeemed

Customers who actively redeem their loyalty points are far less likely to churn. Redemption is a measure of engagement — a customer who is converting points into real rewards clearly values the programme. Customers with zero or near-zero redemption are at elevated churn risk.



4.3 Multivariate Analysis — Feature Relationships

The correlation heatmap reveals several important relationships. CLV is positively correlated with total flights and points accumulated, confirming that frequent flyers generate higher lifetime value. Distance and total flights are highly correlated, which is expected. The churn target shows the strongest negative correlations with flight activity and points accumulated.



4.4 Key EDA Findings



Finding 1 — Flight Activity is the Strongest Churn Signal

Customers who flew less than 5 times per year were disproportionately likely to churn. Flight frequency is the most actionable lever for retention — re-engagement campaigns with flight discounts or bonus points for the next booking directly address the root cause.



Finding 2 — Aurora Members Carry the Highest Individual Risk

Aurora members have the highest average CLV (\$10,673) but also the highest churn rate (13.1%). A single churned Aurora customer represents more lost revenue than two churned Star members. Aurora retention campaigns offer the highest ROI per dollar spent.



Finding 3 — Redemption Gap Signals Disengagement

A significant portion of customers have accumulated thousands of points but have never redeemed any. Unredeemed points indicate customers who enrolled, collected points passively, and then lost interest. Triggered redemption reminders for this segment could re-activate engagement.



Finding 4 — New Members Churn Within 12 Months

The churn rate is significantly higher in the first 12 months of membership. Customers who pass the 12-month threshold and remain active are far more likely to stay long-term. This points to the critical importance of an effective onboarding and early-engagement programme.

5. Feature Engineering

Raw data fields often do not tell the complete story. Feature engineering is the process of creating new, more meaningful variables from existing data — transforming raw numbers into insights that a machine learning model can act on. This step is often what separates a mediocre model from a high-performing one.

The following features were engineered from the integrated dataset:

Engineered Feature	How It Was Created	Why It Matters
Customer Tenure (Months)	Difference between enrollment date and current period	Longer-tenured customers are more loyal and less likely to churn
Avg Flights per Month	Total flights divided by active months	Captures engagement intensity rather than just raw totals
Total Distance Travelled	Sum of all monthly distances across the customer's history	Proxy for overall programme engagement and revenue contribution
Points Redemption Rate	Points Redeemed divided by Points Accumulated	High redemption = active engagement; low redemption = at-risk customer
Engagement Score	Weighted composite: 40% flights + 30% redemption + 30% points normalised	Single summary metric of how engaged a customer is with the programme
Points Utilisation Ratio	Redeemed points as a proportion of maximum possible redemption	Distinguishes passive collectors from active redeemers
Loyalty Efficiency Score	Points Accumulated divided by Distance flown	Identifies customers who earn points efficiently vs those who fly long distances for few points
Salary Group	Binned salary into 5 income bands: <\$40K, \$40-60K, \$60-80K, \$80-100K, \$100K+	Captures non-linear income effects on churn behaviour
Membership Age Category	Grouped tenure into: New (<12m), Growing (12-36m), Established (36m+)	Identifies the high-risk new member cohort that churns early



Why Feature Engineering Matters

A raw column like 'Points Accumulated' tells you how many points a customer has earned. But 'Points Redemption Rate' tells you how actively they are using the programme. The second feature is far more predictive of churn — it captures behaviour, not just quantity. Every engineered feature in this project was designed to capture customer behaviour and engagement that raw data fields cannot express directly.

6. Feature Encoding & Scaling

Machine learning algorithms work with numbers — they cannot directly process text values like 'Aurora' or 'Married'. Before training any model, categorical variables must be converted into numeric form and numeric features must be brought onto a comparable scale.

6.1 Categorical Encoding

Two encoding methods were applied depending on the nature of each categorical variable:

Variable	Encoding Method	Reason
Education	One-Hot Encoding	No natural ordering between education levels (e.g., College is not 'more' than Bachelor)
Marital Status	One-Hot Encoding	Nominal categories with no inherent rank order
Loyalty Card	One-Hot Encoding	Tier labels (Star, Nova, Aurora) are nominal — ordinal encoding could imply incorrect mathematical relationships
Province / City	One-Hot Encoding	Geographic labels with no ordinal meaning
Customer Segment	One-Hot Encoding	Derived segment name, categorical and unordered

6.2 Feature Scaling

Numeric features were scaled using StandardScaler, which transforms each feature to have a mean of 0 and a standard deviation of 1. This is important because without scaling, a feature like 'Distance' (which can be in the hundreds of thousands of kilometres) would dominate a feature like 'Redemption Rate' (which ranges from 0 to 1), simply because of the difference in magnitude — not because it is actually more important.



Why StandardScaler Was Chosen

StandardScaler was chosen over MinMaxScaler because the data contains outliers — some customers fly extremely long distances or earn exceptionally high CLV. MinMaxScaler is sensitive to outliers (a single extreme value compresses all other values into a tiny range). StandardScaler is more robust in this situation.

All preprocessing steps — imputation, encoding, and scaling — were combined into a single sklearn ColumnTransformer pipeline. This ensures that the exact same transformations applied during training are also applied during inference, preventing data leakage and ensuring deployment consistency.

7. Machine Learning Model Development

With clean, engineered, and scaled data in hand, the next step was to build and compare multiple machine learning models. Rather than selecting one algorithm arbitrarily, seven different classifiers were trained and evaluated. This comparative approach ensures the final model is genuinely the best option for this specific problem and dataset.

7.1 Problem Type & Target Variable

This is a binary classification problem. The target variable is Cancelled — 1 if the customer churned (cancelled their loyalty membership), 0 if they remained active. The class distribution is imbalanced: approximately 12.4% of records represent churned customers and 87.6% represent retained customers. This imbalance was addressed using SMOTE (Synthetic Minority Oversampling Technique) to generate synthetic examples of the minority class during training, ensuring the model does not simply learn to predict 'retained' for every customer.

7.2 Train-Test Split

The dataset was divided into a training set (80%) and a held-out test set (20%) using stratified splitting — meaning the 12.4% churn rate was preserved in both subsets. Stratification prevents the situation where the test set accidentally contains very few or very many churned customers, which would produce misleadingly optimistic or pessimistic evaluation results. A fixed random seed (random_state=42) was used to ensure reproducibility.

7.3 Models Trained & Compared

Model	Accuracy	Precision	Recall	F1	ROC-AUC	Selected
Logistic Regression	82.3%	61.4%	69.8%	65.3%	0.831	
Decision Tree	88.7%	71.2%	74.5%	72.8%	0.872	
Random Forest	97.1%	94.5%	81.6%	87.5%	0.981	★ Best
XGBoost	95.8%	91.3%	78.9%	84.7%	0.963	
LightGBM	95.2%	89.7%	77.4%	83.1%	0.958	
Gradient Boosting	94.1%	87.6%	75.2%	80.9%	0.946	
AdaBoost	90.3%	76.8%	72.1%	74.4%	0.908	

7.4 Best Model — Random Forest

The Random Forest classifier was selected as the final model based on its superior performance across every evaluation metric. Here is why each metric matters for this specific business problem:

Metric	Score	Business Meaning
ROC-AUC	0.981	The model is almost perfect at distinguishing churners from non-churners across all decision thresholds
Accuracy	97.1%	97 out of 100 customers are classified correctly
Precision	94.5%	When the model flags a customer as likely to churn, it is correct 94.5% of the time — minimising wasted retention spend
Recall	81.6%	The model successfully identifies 82% of customers who will actually churn — limiting missed at-risk customers
F1 Score	87.5%	Strong balance between Precision and Recall — the model is both accurate and comprehensive

Random Forest was selected over XGBoost (ROC-AUC 0.963) because of its higher precision (94.5% vs 91.3%). In a retention context, high precision is critically important — sending a retention offer to 1,000 customers who are not actually at risk wastes budget and can feel intrusive. Random Forest's higher precision means the retention team can be confident that flagged customers are genuinely at risk.

8. Model Performance & Business Insights

8.1 Risk Segmentation — All Customers Scored

The trained model was applied to all 16,737 unique customers in the loyalty database to generate an individual churn probability score for each. Customers were then classified into three risk tiers based on their probability:

Risk Category	Probability Range	Customer Count	% of Base	Recommended Action
Low Risk	0% – 40%	7,691	46.0%	Standard communications — maintain engagement
Medium Risk	40% – 70%	6,627	39.6%	Targeted re-engagement — double points, discount offers
High Risk	70% – 100%	2,419	14.5%	Immediate personalised retention intervention

8.2 Top Predictive Features

Rank	Feature	Importance	What It Means for the Business
1	Enrollment Year	0.116	Newer members churn at higher rates — onboarding is a critical retention window
2	Membership Age (Months)	0.079	Tenure is strongly protective — loyalty grows with time in the programme

3	Total Flights	0.073	Flight activity is the clearest signal of programme engagement
4	Tenure	0.070	Long-standing customers are deeply embedded in the loyalty ecosystem
5	CLV	0.051	Higher lifetime value correlates with lower churn — valuable customers stay

8.3 Strategic Retention Recommendations

- **Customers with both high churn probability and high CLV represent the greatest financial risk. A personalised retention offer — tier upgrade, bonus points, or exclusive benefit — to this group delivers the highest ROI per retention dollar:** Target the High-Risk + High-CLV Quadrant First
- **Since Enrollment Year is the strongest churn predictor, new members must be actively engaged in their first 90 days. Automated welcome emails, first-flight incentives, and milestone rewards can significantly reduce early churn:** Build a 90-Day New Member Programme
- **Customers with large unredeemed balances who have not flown recently are at elevated risk. Triggering a 'Your points are about to expire' notification with a redemption incentive is a proven re-engagement tactic:** Reactivate Dormant Point Holders
- **Aurora members churn for different reasons than Star members — they expect premium experiences and personalised service. A dedicated Aurora concierge programme or exclusive travel benefits can address the expectation gap:** Differentiate Aurora Retention Strategy
- **The predictive dashboard provides a live risk score for every customer. Customer success teams should run a monthly review of the High Risk table and action the top 200 customers each month:** Use the Dashboard for Monthly Risk Reviews

9. Dashboard Development

Building a great machine learning model is only half the job. The real business value is unlocked when that model is made accessible to the people who can act on it — marketing managers, customer success teams, and business executives who do not have data science training. This is exactly what the dashboard was built to do.

The dashboard was developed using Streamlit — a Python framework that allows data scientists to build interactive web applications without frontend development experience. The result is a five-page web application that runs in any browser and requires no technical knowledge to use.

9.1 Page 1 — Executive Dashboard (Home)

The home page is designed for senior stakeholders who need a quick overview. It displays six headline KPI cards showing total customers, average CLV, churn rate, total flight activity, active members, and overall revenue indicators. Four auto-generated executive highlights summarise the most important findings from the current data.

9.2 Page 2 — Customer Analytics

This page provides deep-dive customer analytics with six interactive sidebar filters — allowing users to slice the customer base by loyalty tier, gender, education level, province, CLV range, and salary range. All charts update in real time when filters are applied. Visualisations include demographics breakdowns, salary distributions by tier, a geographic activity heatmap, and a behavioural scatter plot comparing CLV against total flights.

9.3 Page 3 — Loyalty Programme Health

The loyalty page provides operational metrics for the programme management team — points accumulated vs redeemed by tier, redemption rate trends, churn rate by tier, a customer flow Sankey diagram (showing how different education levels flow into different loyalty tiers and churn/retention outcomes), and a monthly points accumulation timeline. Five data-driven recommendations are auto-generated from current programme metrics.

9.4 Page 4 — Revenue Insights

The revenue page provides financial decomposition analysis — a revenue waterfall by loyalty tier, Pareto analysis of cumulative revenue concentration, a province-level revenue treemap, and revenue decomposition separating CLV-based contributions from points redemption costs.

9.5 Page 5 — Predictive Analytics (Churn Prediction)

This is the most technically sophisticated page — and the most directly actionable. It integrates the trained Random Forest model directly into the dashboard with the following features:

- Individual Customer Predictor — a searchable dropdown to select any of the 16,737 customers by their Loyalty Number, instantly displaying their personalised churn probability
 - Plotly Gauge Chart — a visual semi-circular gauge showing the churn probability with green (Low Risk), orange (Medium Risk), and red (High Risk) colour zones
 - Risk Badge — a colour-coded pill label (Low / Medium / High Risk) with an emoji indicator
 - Customer Summary Card — displays CLV, total flights, loyalty tier, revenue contribution, and risk level for the selected customer
 - Auto-Generated Retention Insight — a plain-English recommendation is automatically generated based on the customer's risk level and profile
 - High-Risk Table — a filterable, sortable table of all customers by risk category, with a one-click CSV download
 - Churn Risk x CLV Quadrant Scatter — visualises every customer plotted by their churn probability and CLV, with annotation lines marking the High Risk and Median CLV thresholds
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10. Project Methodology & Architecture

The following diagram illustrates the end-to-end flow of the project — from raw data to deployed dashboard. Each stage fed directly into the next, with clean handoffs between data engineering, machine learning, and application development.

1	Data Collection	Three datasets collected: Customer Loyalty History, Flight Activity, and Calendar data
2	Data Cleaning	Missing value treatment, duplicate removal, data type corrections, and data integrity validation
3	Data Preprocessing	Dataset merging to create unified analytical table; train-test split (80/20 stratified)
4	EDA	Univariate, bivariate, and multivariate analysis to discover churn patterns and key drivers
5	Feature Engineering	9 new features created: Tenure, Avg Flights/Month, Redemption Rate, Engagement Score, and more
6	Encoding & Scaling	One-Hot Encoding for categoricals; StandardScaler for numerics inside ColumnTransformer pipeline
7	Model Training	7 algorithms trained: Logistic Regression, Decision Tree, Random Forest, XGBoost, LightGBM, Gradient Boosting, AdaBoost
8	Model Evaluation	Compared on Accuracy, Precision, Recall, F1, ROC-AUC, and Confusion Matrix on held-out test set
9	Best Model Selection	Random Forest selected — highest ROC-AUC (0.981), Accuracy (97.1%), and Precision (94.5%)
10	Artefact Persistence	Model saved to models/churn_model.pkl via Joblib — loaded at startup, no retraining required
11	Dashboard Development	5-page Streamlit application with all 16,737 customers scored and interactive visualisations built
12	Deployment	Deployed to Streamlit Community Cloud via GitHub — accessible from any browser, no installation needed

11. Conclusion

This project successfully demonstrates what an end-to-end data science solution looks like in practice — from a clearly defined business problem through rigorous data preparation, exploratory analysis, machine learning, and finally a deployed application that real business users can access and benefit from.

The churn prediction model achieves a ROC-AUC of 0.981 and 97.1% accuracy — performance that is both statistically rigorous and practically meaningful. The model identifies 2,419 customers who are at high risk of leaving the loyalty programme, giving the airline's retention team a prioritised, data-driven list to act on immediately. With an average Aurora-tier CLV of \$10,673, retaining even 20% of high-risk Aurora customers represents a significant revenue recovery.

The five-page Streamlit dashboard transforms the model's predictions into an accessible business intelligence application. A marketing manager can open the dashboard, select any customer by their loyalty number, and immediately see that customer's churn probability, risk category, CLV, and a personalised retention recommendation — all without writing a single line of code.



Key Takeaway

This project shows that data science creates the most value when it connects the technical and the practical — when a great model is paired with an accessible interface that empowers business teams to act on its outputs. The airline now has a tool that can turn churn prediction from a quarterly batch exercise into a continuous, real-time retention operation.

Links

GitHub Repository: <https://github.com/vnareshraiput/Airline-Loyalty-Customer-Churn-Prediction.git>

Streamlit Dashboard: <https://airline-loyalt-churn-prediction.streamlit.app/>

Contact: naresh_v@bt.iitr.ac.in

— End of Report —